

A SATELLITE-BASED METHODOLOGY FOR HARVEST DATE DETECTION AND YIELD PREDICTION IN SUGARCANE

Yuri Shendryk¹, Lecheng Pan¹, Matthew Craigie¹, Mattia Stasolla², Catherine Ticehurst³, and Peter Thorburn¹

¹CSIRO, Agriculture and Food, Australia

²Signal and Image Centre, Royal Military Academy, Belgium

³CSIRO, Land and Water, Australia

ABSTRACT

An accurate model of yield prediction will benefit many aspects of managing growth and harvest of sugarcane crops. In this study Sentinel-1 and Sentinel-2 time-series were used to automatically detect harvest dates of sugarcane fields in the Far North Queensland of Australia. Harvest date information was further used in combination with weather, soil and elevation data to predict sugarcane yield at different time steps over three consecutive growing seasons using machine learning. Our results suggest that harvest dates could be identified with detection rates of 87% and 91% using Sentinel-1 and Sentinel-2 imagery, respectively. Similarly, sugarcane yield could be predicted using Sentinel-1 and Sentinel-2 satellite imagery in conjunction with other geographical attributes with accuracy of 65% as early as 180 days after the previous harvest.

Index Terms — sugarcane, yield, harvest date, satellite, multispectral, SAR, yield, fusion, time-series, Sentinel-1, Sentinel-2

1. INTRODUCTION

In Australia around 30.5 million tons of sugarcane is harvested and processed each growing season, producing around \$1.2 billion in revenue [1]. There are multiple stakeholders across the agricultural value chain interested in understanding and forecasting supply in the sugarcane market, including sugarcane growers, sugar mills, commodities traders, banks and insurance companies as well as biofuel producers. The efficient management of the fertilization and harvesting processes also helps to increase profitability in the sugarcane sector. Therefore, an accurate model for sugarcane yield prediction is needed to manage sugar mills and allocate other stakeholders' resources. Additionally, knowing yield predictions at early stages

of crop development allows sugarcane growers to reduce excessive fertilizer application, which has the combined benefits of improving farm profitability and reducing nitrogen discharge into catchments [2].

Sugarcane supply can vary according to the cultivation area, climatic conditions and disease. Unfortunately, in most countries, sugarcane yield predictions are often still performed through traditional methods that are site-specific and time-consuming. In the traditional system, sugarcane yield is estimated based on the planting records or allometric models. These approaches often require visual interpretation by an experienced eye, and farmers' knowledge on historical production and environmental conditions [3]. Therefore, remote sensing has recently been proposed as an alternative approach to predict sugarcane yield.

The majority of remote sensing studies have used satellite imagery to predict sugarcane yield, but these have had mixed success. To date, such studies have been relatively limited in their scope; often only detecting reasonably strong correlations between satellite imagery and yield of sugarcane when considering yield and imagery averaged over large regions [4]. Other studies have developed strong models for yield estimation at the field level, but used only a small sample of data collected as part of field experiments [5], while others found only limited correlations at the field level using data provided by sugar mills [6, 7]. Multiple studies have also indicated that the success of sugarcane yield prediction depends on time-series analysis [4-6], for which accurate harvest date detection is necessary.

The aim of this study was to see if useful models for predicting sugarcane yield at the field level could be developed from freely available Sentinel-1 and Sentinel-2 satellite imagery. Since harvest date is an important input for yield predictions, the accuracy of

harvest date detection at the field level using both Sentinel-1 and -2 imagery was also evaluated.

2. METHODOLOGY

2.1. Study area and field data

Field boundaries, harvest dates and corresponding yield data were provided by farmers in the Far North Queensland (Figure 1) for growing seasons of 2016, 2017 and 2018. The information was provided for the total number of 1929 sugarcane fields, which were standardized to account for changing boundaries between growing seasons. Yield information was also reclassified into ‘*high yield*’ (≥ 110 t/ha), ‘*medium yield*’ (>80 and <110 t/ha) and ‘*low yield*’ (≤ 80 t/ha) classes using the Jenks natural breaks classification method [8].

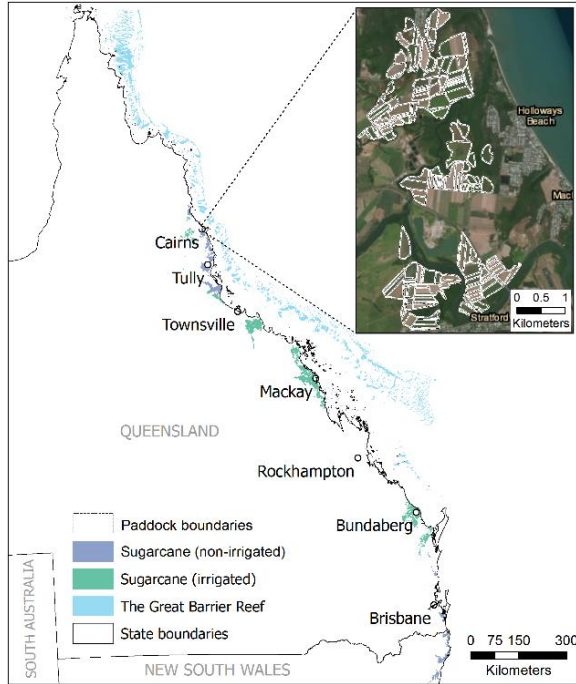


Figure 1. An example of sugarcane fields used in this study in relation to sugarcane cultivation areas of Australia.

2.2. Satellite imagery

The satellite imagery consisted of Sentinel-1 SAR and Sentinel-2 multispectral imagery covering the area of the Far North Queensland at 20 m resolution every 12 days and 10 m resolution every 5 days, respectively. Both Sentinel-1 and Sentinel-2 data were downloaded in the period from 01/12/2015 to 01/12/2018 to cover three growing seasons and processed using SNAP and Sen2Cor software. Both VH and VV polarization Sentinel-1 imagery were noise-corrected, orthorectified and calibrated to Sigma0 backscatter coefficients. Similarly, Sentinel-2 imagery was processed to surface

reflectance, cloud and cloud shadow masked, resampled to 10 m spatial resolution, and further used to calculate 10 vegetation indices (VI) (Table 1).

Table 1. Combinations of Sentinel-2 bands to estimate multiple vegetation indices (B – blue at 490 nm, G – green at 560 nm, R – red at 665 nm, RE1 – red edge at 705 nm, RE2 – red edge at 740 nm, NIR1 – near-infrared at 842 nm, NIR2 – near-infrared at 865 nm, SWIR1 – short-wave infrared at 1610 nm, SWIR2 – short-wave infrared at 2190 nm).

VI	Equation
NDVI1	$(\text{NIR1} - \text{Red}) / (\text{NIR1} + \text{Red})$
GNDVI	$(\text{NIR1} - \text{Green}) / (\text{NIR1} + \text{Green})$
NDWI	$(\text{NIR1} - \text{SWIR1}) / (\text{NIR1} + \text{SWIR1})$
NDTI	$(\text{SWIR1} - \text{SWIR2}) / (\text{SWIR1} + \text{SWIR2})$
NDVI2	$(\text{RE2} - \text{SWIR2}) / (\text{RE2} + \text{SWIR2})$
NDVI3	$(\text{NIR2} - \text{RE1}) / (\text{NIR2} + \text{RE1})$
EVI	$2.5 \times (\text{NIR1} - \text{Red}) / ((\text{NIR1} + 6.0 \times \text{Red} - 7.5 \times \text{Red}) + 1.0)$
SAVI	$(\text{NIR1} - \text{Red}) / (\text{NIR1} + \text{Red} + 0.5) \times (1.0 + 0.5)$
ARVI	$(\text{NIR1} - \text{Red} - 0.106 \times (\text{Red} - \text{Blue})) / (\text{NIR1} + \text{Red} - 0.106 \times (\text{Red} - \text{Blue}))$
MCARI	$((\text{RE1} - \text{Red}) - 0.2 \times (\text{RE1} - \text{Green})) \times (\text{RE1} / \text{Red})$

Finally, time-series from both Sentinel-1 (VH and VV backscatter) imagery and Sentinel-2 derived VIs were calculated by extracting statistics for each of the sugarcane fields (Table 2) resulting in 14 and 70 SAR and multispectral-derived time-series.

Table 2. Statistics extracted from Sentinel-1 and Sentinel-2 imagery.

Statistics	Description
min	The minimum value of all pixels within a field
max	The maximum value of all pixels within a field
avg	The average value of all pixels within a field
std	The standard deviation of all pixels within a field
p25	The 25 th percentile of all pixel values within a field
p50	The 50 th percentile of all pixel values within a field
p75	The 75 th percentile of all pixel values within a field

2.3. Harvest date detection

In this study the ability of both Sentinel-1 and Sentinel-2 imagery in detecting harvest dates was compared. To make harvest date detection results comparable 125 sugarcane fields that had their boundaries unchanged in 2017 and 2018 (i.e. 250 harvest dates in total) were selected. To determine the quality of detections two accuracy metrics were used: 1) detection rate – the rate of correctly detected harvest dates (matching the ground truths within 24 days) and 2) false detection rate – rate of the detections that did not correspond to true harvest dates (within 24 days).

2.3.1. Sentinel-1

Harvest date detection using Sentinel-1 imagery was based on time-series of average VH backscatter values within field boundaries, and was performed using a

change point detection technique [9]. In order to reduce the effect of the speckle noise, the Sentinel-1 time series have been preliminarily smoothed by applying a Gaussian filter with a standard deviation of 10.

2.3.2. Sentinel-2

Harvest date detection using Sentinel-2 imagery was based on a thresholding approach. Passing through the time-series, when the value of a VI dropped below the threshold value, a harvest date was detected. To find an optimal threshold an exhaustive grid search over different Sentinel-2 VIs (Table 1), with different thresholds and subsequent observations (i.e. 1, 2 and 3) below the threshold was performed. Sugarcane fields that contained less than 10 Sentinel-2 observations were excluded from the harvest detection procedure.

2.4. Yield prediction

Using the Sentinel-2 based harvest date detection method described above, Sentinel-1 VV and VH backscatter as well as Sentinel-2 VIs time-series for growing seasons of 2016, 2017 and 2018 years within 1929 sugarcane fields were extracted. Sugarcane fields that contained less than 10 Sentinel-2 observations or less than 5 observations with $NDVI > 0.6$ were excluded from further analysis. Then both Sentinel-1 and Sentinel-2 time-series were converted to daily time-series using spline interpolation. Sentinel-2 VIs were also smoothed using the Savitzky-Golay filter with 91 days filter window and 2nd order polynomial to fit the data. Out of 1929 sugarcane fields only 761 remained for yield prediction.

In addition to satellite based observations, airborne LiDAR-derived elevation, soil type and weather information were incorporated into the analysis to predict sugarcane yield at the field level. Predictor variables from the elevation layer were calculated by extracting statistics for each of the sugarcane fields described in Table 2. Similarly the most dominant soil type within a sugarcane field was used as a predictor variable. In contrast, daily maximum and minimum daily temperatures, precipitation, radiation and evaporation time-series were extracted from the closest meteorological station to each sugarcane field and clipped to the length of the growing season in each year. Predictor variables from weather time-series were calculated by extracting statistics for each of the sugarcane fields described in Table 3. Similarly, in order to capture most of the signal in the Sentinel-1 and Sentinel-2 time-series, the predictor variables described in Table 3 were also extracted.

Table 3. Predictor variables extracted from Sentinel-1 VV and VH backscatter, Sentinel-2 VIs and weather time-series.

Name	Description
count above mean	the number of values in time-series that are higher than the mean of time-series
count below mean	the number of values in time-series that are lower than the mean of time-series
kurtosis	the kurtosis of time-series
max	the maximum value of time-series
mean	the mean value of time-series
median	the median value of time-series
min	the minimum value of time-series
skewness	the skewness of time-series
std	the standard deviation of time-series
mean abs change	the mean over the absolute differences between subsequent time-series values
mean change	the mean over the differences between subsequent time-series values

In total, there were 154, 770, 7, 1, and 55 predictor variables, extracted from Sentinel-1 VV and VH backscatter, Sentinel-2 VIs, elevation, soil and weather data, respectively. To evaluate the predictive ability at different time steps, the abovementioned predictor variables were extracted at 30, 60, 90, 120, 150 and 180 days after the previous harvest date (DAH).

The extreme gradient boosting decision tree (XGBoost) algorithm [10] was further used to predict sugarcane yield at the field level. Sequential forward feature selection based on predictor variable importance was also applied to reduce the dataset dimensionality and improve the performance of the predictive models. Finally, balanced accuracy (BA), calculated as the average of a percentage of correctly classified samples within each class (i.e. ‘high’, ‘medium’ and ‘low’ yield), was used to evaluate model performance at training (75% of the data) and validation (25% of the data) stages.

3. RESULTS AND DISCUSSION

3.1. Harvest date detection

As reported in Table 4, the analysis of the Sentinel-1 time-series allowed us to achieve a detection rate of 86.8%. The number of false alarms was 37, resulting in a false detection rate of 1%. Similarly, the best combination of parameters in Sentinel-2 based harvest date detection approach was using the median ($p50$ in Table 2) NDVI value and detecting a harvest date when two subsequent observations had values below a threshold of 0.42. Using this method, a detection rate of up to 91.2% was achieved with a false detection rate of 1% (Table 4).

Table 4. The accuracy of harvest date detection (excluding fields affected by cloud and cloud shadow masking, i.e. with <10 observations per growing season).

	Sentinel-1	Sentinel-2
True harvest dates	250	250
Missed harvest detections due to cloud and cloud shadow masking	0	48
Detected harvest dates	217	184
True detection rate	86.8%	73.6% (91.2% ¹)
False positive rate	1%	1%

Since the Sentinel-1 signal is able to penetrate clouds, harvest dates in all 250 sugarcane fields were identified, while Sentinel-2 cloud cover prevented the detection of harvest dates in 48 sugarcane fields. Thus, the harvest date detection rate using Sentinel-2 imagery was 73.6% and 91.2% when including and excluding fields affected by cloud and cloud shadow masking into accuracy metric calculation, respectively. An example of the harvest date detection using Sentinel-2 derived median NDVI time-series is shown in Figure 2.

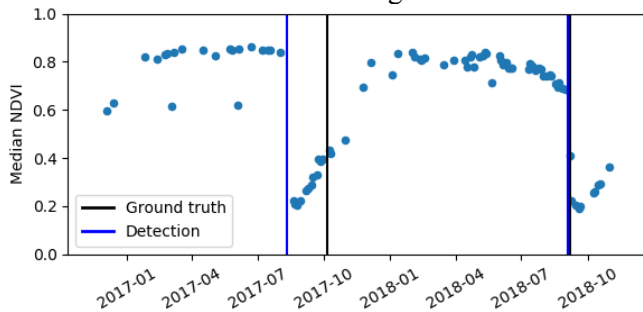


Figure 2. Time-series of Sentinel-2 derived median NDVI over an arbitrary sugarcane field with automatically detected harvest dates.

3.2. Yield prediction

Overall, the performance of yield predictions peaked at around 150 to 180 DAH when the crop was starting to mature. The combined predictors based model performed better (BA of 64.7%) than the satellite imagery based model (BA of 55.2%) and the model based on NDVI, soil type, elevation, and weather information (BA of 58.3%) (Table 5).

Table 5. Yield prediction accuracy at different DAH in terms of balanced accuracy (BA, %). A BA of 33% corresponds with a random guess.

DAH	Satellite imagery predictor variables	NDVI, soil type, elevation, and weather predictor variables	All predictor variables combined
30	45.5	52.9	46.0
60	50.6	49.8	52.6
90	52.8	54.6	56.3
120	53.9	51.1	62.9
150	55.2	58.3	61.5
180	54.7	56.9	64.7

The application of a previously reported method suggesting regressing the average GNDVI (Table 1) value over the growing season against sugarcane yield (R^2 of 0.43 [7]) in this study resulted in R^2 of only 0.1.

4. CONCLUSION

This study demonstrated the potential of using Sentinel-1 and Sentinel-2 satellite imagery in conjunction with other geographical attributes for predicting sugarcane yield at the field level. Our models for detecting harvest dates achieved an accuracy of up to 91.2%. It was also found that performance of yield predictions peaked around 180 DAH when using the fusion of satellite imagery, elevation, soil type and weather information (BA of 64.7%). Our models could be used to predict yield in subsequent growing seasons and thereby inform nutrient management and harvest programs.

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